

GMVAE-4P: Multi-Modal Deconvolution for Patient Phenotyping from Bulk RNA-seq via Uncertainty-Aware Transfer Learning

ABSTRACT

GMVAE-4P trains on scRNA-seq but *deploys on bulk RNA-seq* (\$100 vs \$10,000 per patient) by recovering three cellular modalities and fusing them under uncertainty-aware attention (**Fig. 1**). Across **834,000 cells from 169 patients** (Lupus, COVID-19, Cardiovascular), it reaches 0.95–0.99 ROC-AUC on scRNA (**Fig. 2**) and 0.87–0.91 on bulk (**Fig. 3**) — 92–95% retention and an 8–16% gain over the strongest baseline.

INTRODUCTION

Bulk RNA-seq is **100× cheaper** than single-cell but averages expression across millions of cells — masking the cellular heterogeneity (**Fig. 7**) that drives immune dysregulation and treatment response. Existing methods force a choice: scRNA classifiers (ProtoCell4P, PaSCient, ScRAT) resolve cells but cost too much; deconvolution tools (CIBERSORT, MuSiC) recover proportions but cannot classify. **None propagate deconvolution uncertainty** downstream. **GMVAE-4P** bridges the gap with cross-resolution transfer learning.

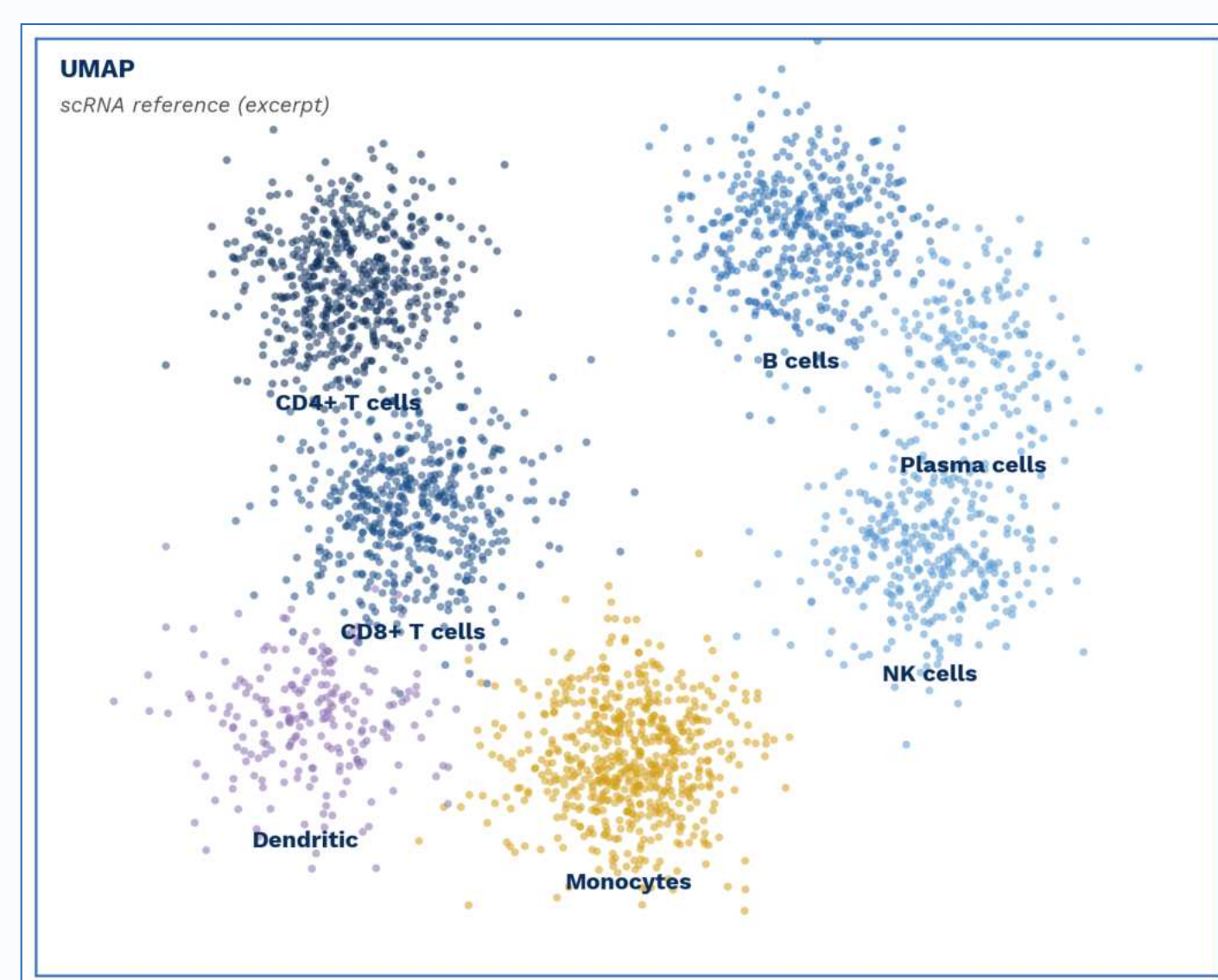


Fig. 7. UMAP of the scRNA reference — clean cell-type clusters are the signal bulk RNA-seq destroys.

BY THE NUMBERS

834K single cells
169 patients
3 diseases
2,021 L-R pairs
Lupus | COVID-19 | Cardiovascular cohorts
100 bootstrap iterations propagate uncertainty

METHODOLOGY

Three complementary modalities are extracted from bulk RNA-seq, each paired with an uncertainty estimate; a learned attention layer fuses them with weights inversely proportional to noise (**Fig. 1**).

M1 — Proportions. *Who?* NNLS deconvolution against a 500-marker scRNA signature.

M2 — Interactions. *Doing?* 2,021 CellChat ligand–receptor pairs across 229 pathways.

M3 — Expression. *How?* Cell-aware z-scoring isolates activation from abundance.

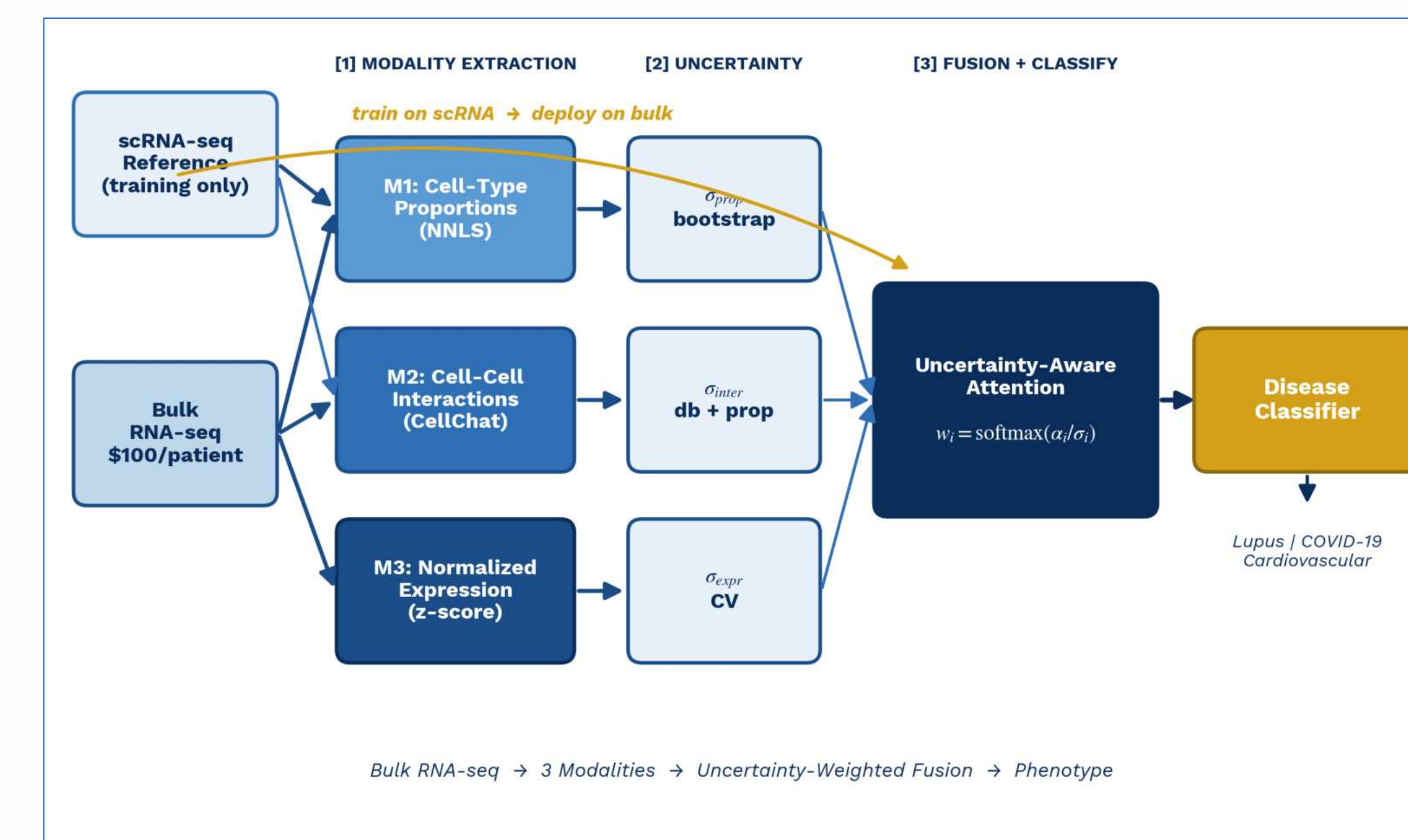


Fig. 1. Three-modality pipeline. Train on scRNA-seq, deploy on bulk. Each modality emits an uncertainty score that down-weights noise in fusion.

KEY EQUATIONS

Deconvolution (per patient):

$$B = S \cdot \Pi + \epsilon$$

bulk = signature × proportions + noise

Uncertainty-aware attention:

$$w_i = \text{softmax}(\alpha_i / \sigma_i)$$

high uncertainty → low attention weight

WHY UNCERTAINTY-AWARE?

Deconvolution noise propagates downstream when modalities aren't weighted by reliability. **Three per-modality reliability scores** let attention down-weight noisy evidence — alone worth 3.4% AUC (**Fig. 4**).

RESULTS

GMVAE-4P beats every scRNA baseline across all three diseases (**Fig. 2**) — an **8–16% AUC gain** over ProtoCell4P, the next-best method.

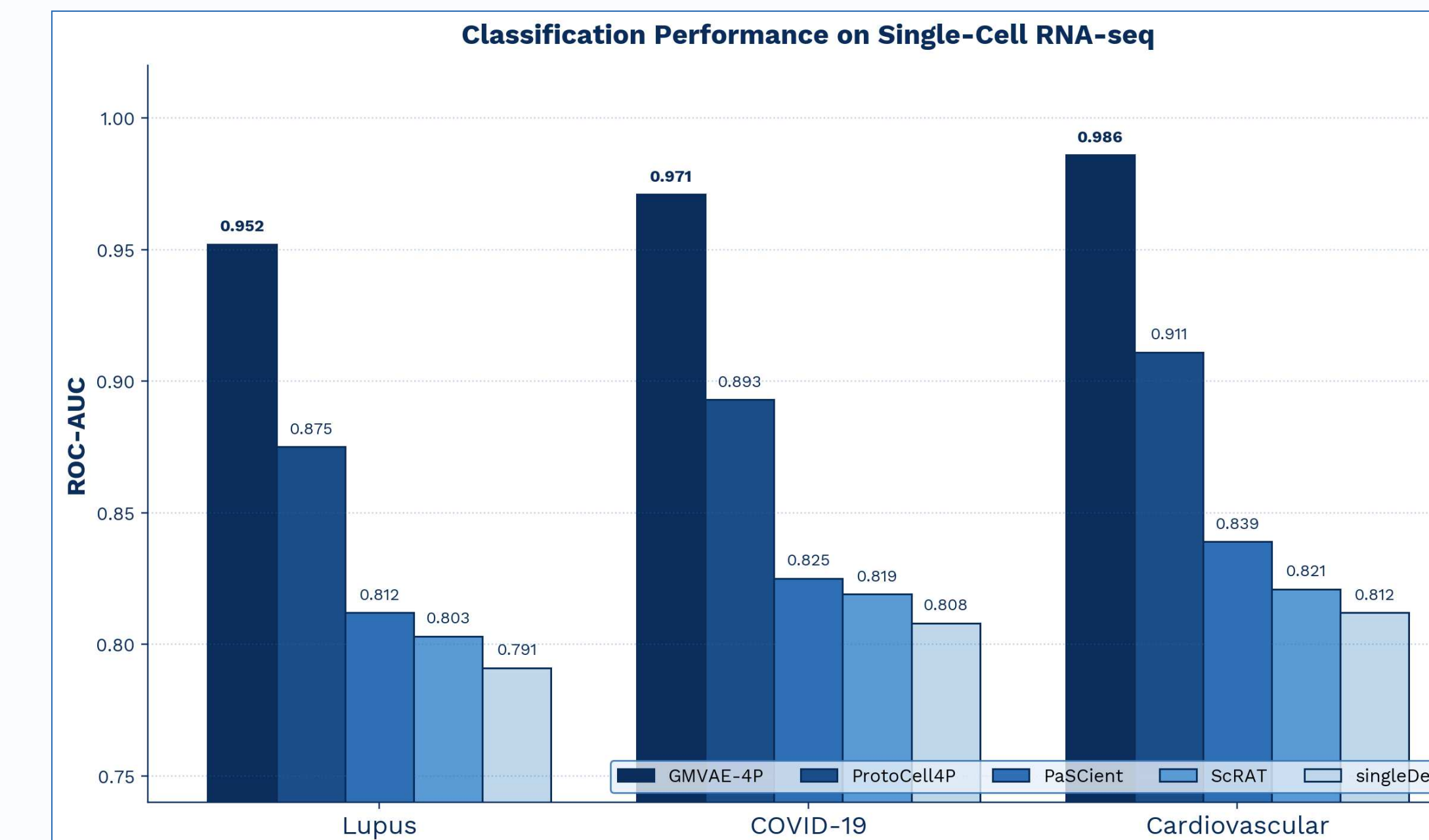


Fig. 2. ROC-AUC on scRNA-seq across three diseases (834K cells, 169 patients).

Transferring weights to bulk RNA-seq retains **92–95% of performance** (**Fig. 3**; $p < 0.001$ across diseases).

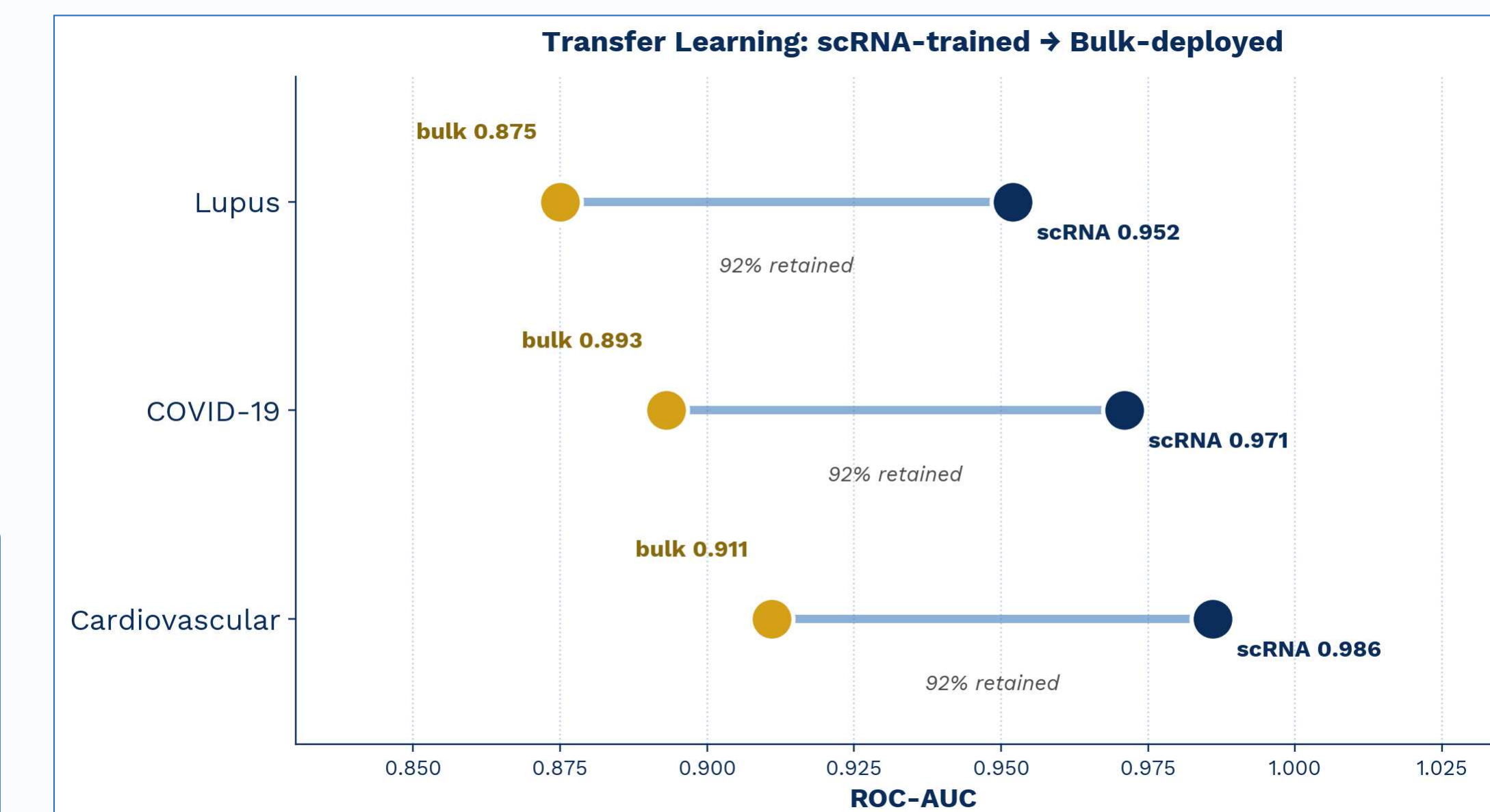


Fig. 3. Transfer-learning retention. Navy: scRNA AUC. Gold: bulk AUC.

Each modality contributes 11–16% AUC; uncertainty weighting adds 3.4% (**Fig. 4**).

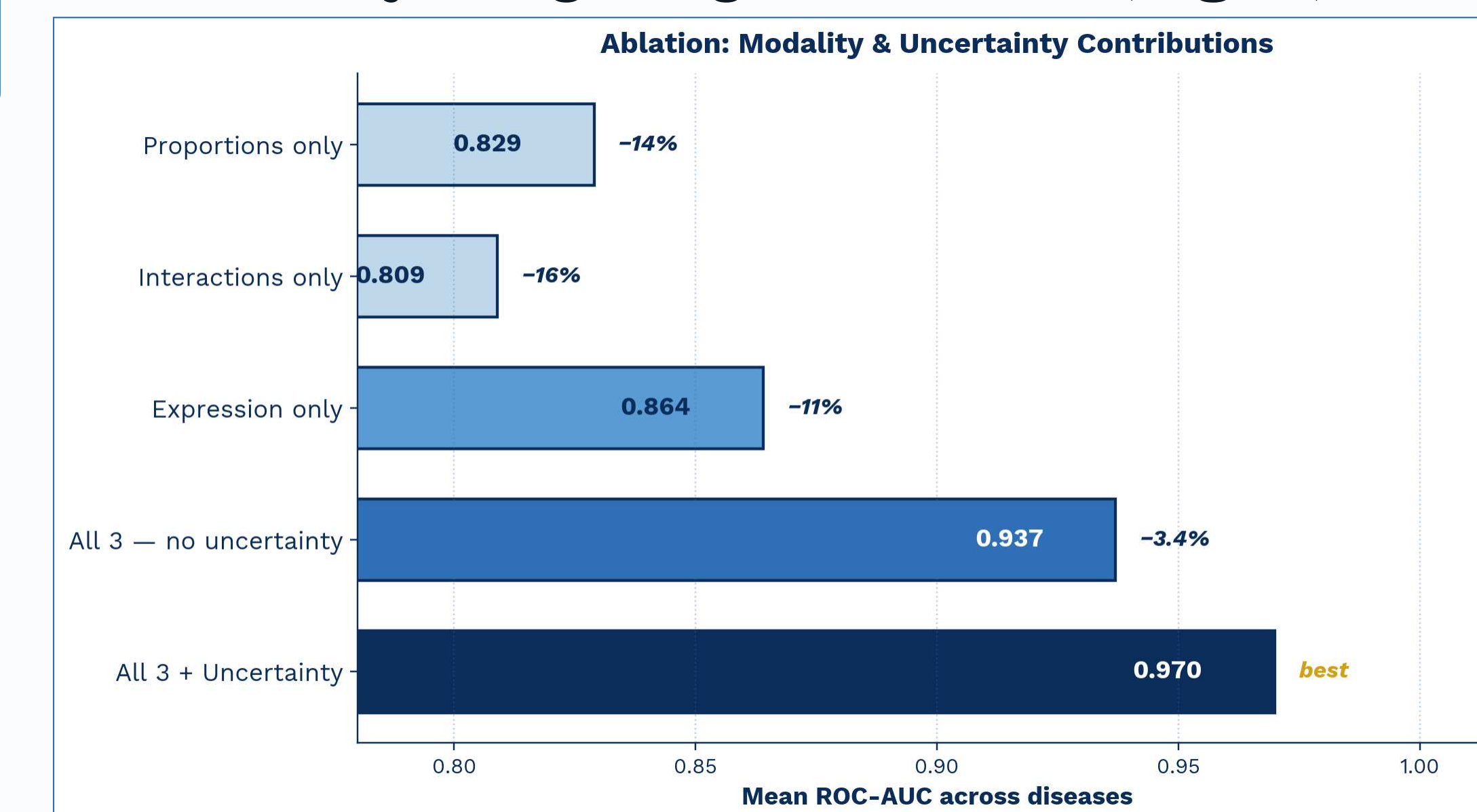


Fig. 4. Ablation. All components contribute independently to the 0.970 best AUC.

RESULTS, cont.

GMVAE-4P delivers a **99% cost reduction** (\$10K→\$100), **60% fewer samples** (200→80), and shortens turnaround from 4–6 weeks to 1–2 (**Fig. 5**) — while retaining 89% of full-resolution AUC.

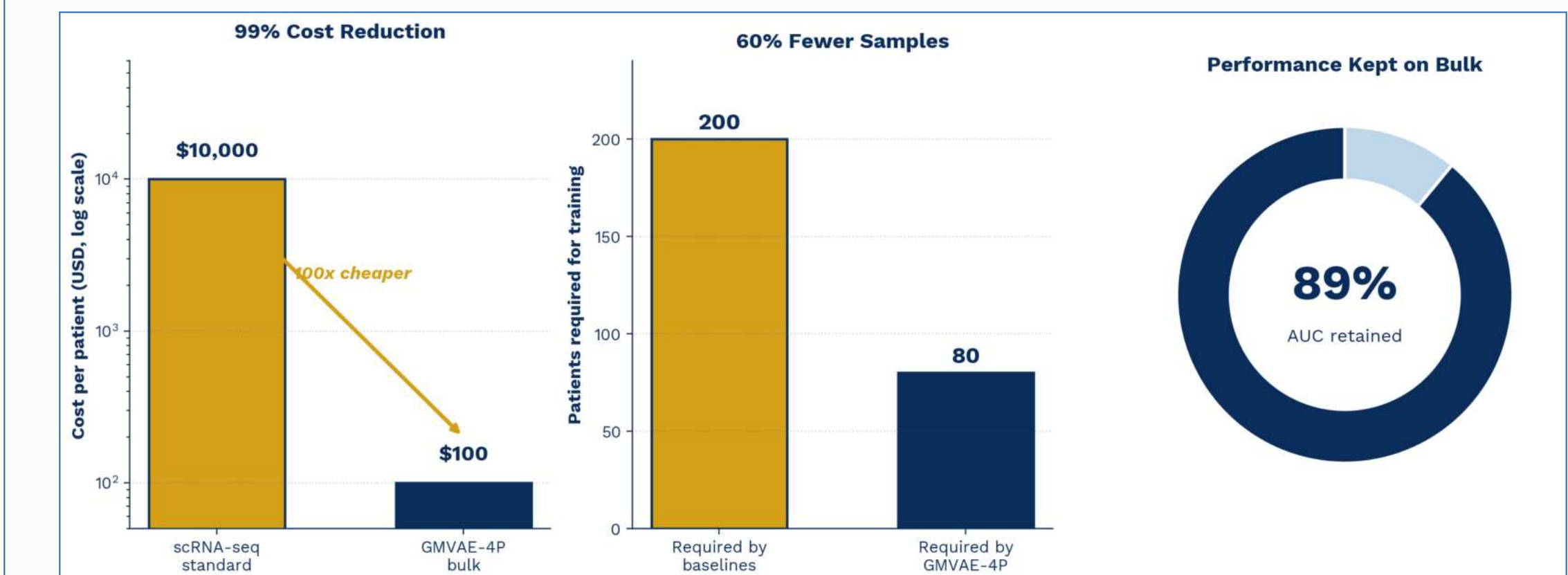


Fig. 5. Deployment metrics — cost, samples, performance retained.

On bulk data, attention shifts toward normalized expression (**Fig. 6**) — the network learns to *trust measurement over deconvolution*.

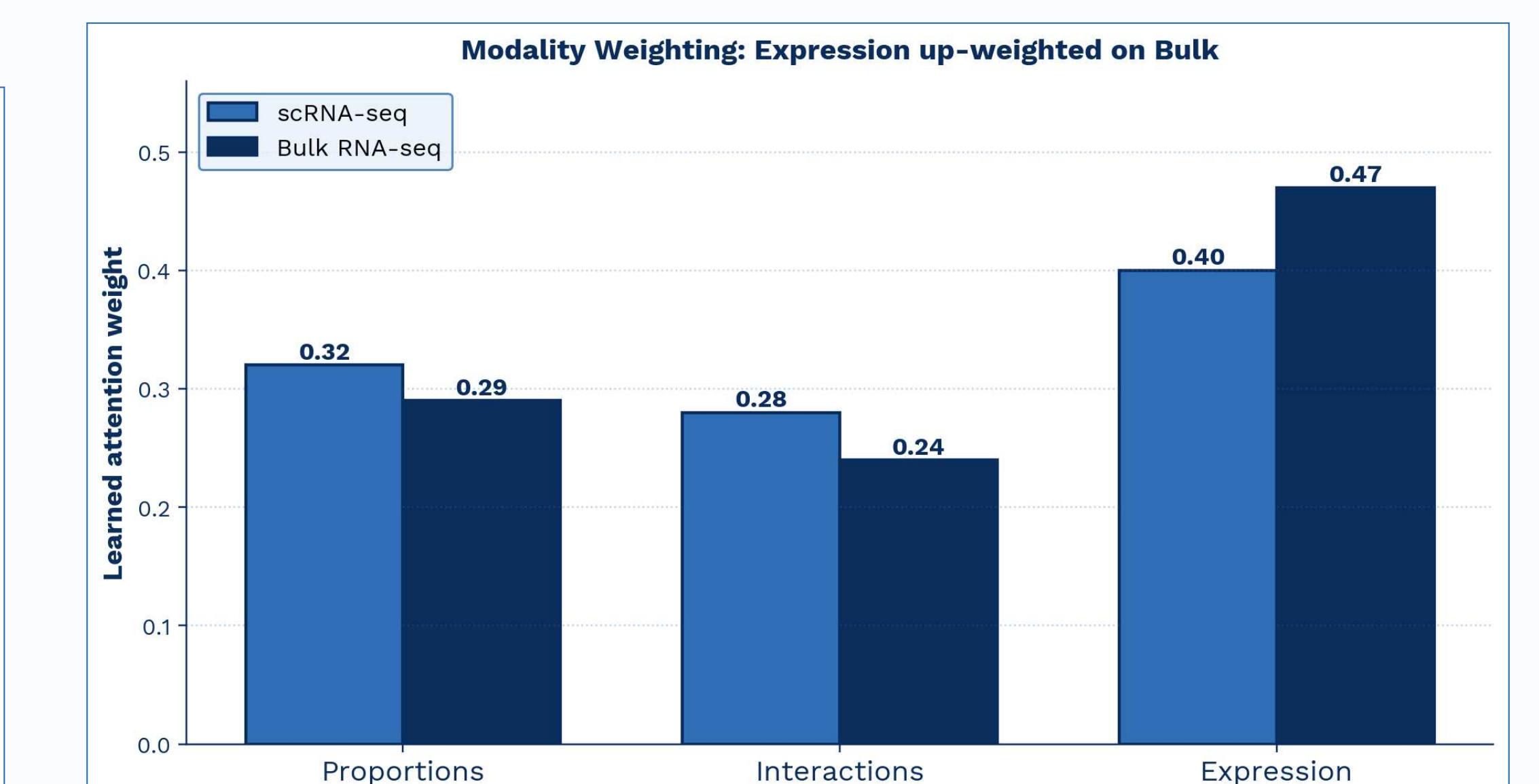


Fig. 6. Learned attention weights — expression up-weighted on bulk.

CONCLUSIONS

- **State-of-the-art on scRNA-seq:** 0.970 mean ROC-AUC (**Fig. 2**).
- **First viable bulk-RNA-seq phenotyping method:** 0.893 mean AUC, 92% retention (**Fig. 3**).
- **Clinical impact:** 100× cheaper, 60% smaller cohorts (**Fig. 5**); enables retrospective TCGA / GTEx mining and community-hospital deployment.
- **Next:** flow-cytometry-constrained deconvolution; cancer & neurodegenerative cohorts; spatial transcriptomics as 4th modality.

REFERENCES

- [1] Newman, A. M. et al. (2019). Determining cell-type abundance and expression from bulk tissues with digital cytometry. *Nature Biotechnology*, 37(7), 773–782. *CIBERSORTx — deconvolution backbone for Modality 1.*
- [2] Jin, S. et al. (2021). Inference and analysis of cell–cell communication using CellChat. *Nature Communications*, 12, 1088. *Ligand–receptor database underpinning Modality 2.*
- [3] Xiong, D. et al. (2023). ProtoCell4P: explainable patient classification from single-cell RNA-seq using prototypical networks. *Bioinformatics*, 39(5), btac112. *Strongest scRNA baseline benchmarked against in Fig. 2.*